Weighted-A^* Based Energy Efficient Resource Allocation in G.Fast

Ankita Raj and Pravesh Biyani

***Abstract*— One of the key challenges in G.fast is to minimize the power consumption at distribution points. G.fast standards define Discontinuous Operation modes that provide avenues for power reduction by allowing intermittent transmission of users along time slots. In this paper, we formulate a power efficiency problem as a user-slot assignment problem, where we schedule users to time slots during discontinuous operation such that the total power consumption is minimized. Since the general user-slot assignment is NP hard, we propose a weighted A^* algorithm based solution that achieves reasonable performance with limited computational resources. Further, the proposed user-slot assignment is also G.fast standards compliant and therefore can be implemented in practice. We also explore different precoding as well as user grouping strategies that can be employed while performing user-slot allocation. Finally, the main insight of this work is that on using our user-slot assignment algorithm and by only precoding during the normal operations, we achieve energy efficiency levels comparable to those achieved when precoding is applied during the discontinuous operations without the suggested user-slot allocation.**

***Index Terms*— DSL, resource management, heuristic algorithms, energy efficiency, precoding.**

I. Introduction

WITH the ever increasing demand for higher data rates and the limited capabilities of traditional copper-based networks, hybrid copper-fiber access networks have emerged as a promising broadband access technology. G.fast is an emerging broadband standard approved by the International Telecommunication Union (ITU), tailored specifically for Fiber to the Distribution Point (FTTdp) deployment, wherein optical fiber from the central office terminates in a distribution point unit (DPU) close to the end-users, and the remaining distance is covered by existing copper infrastructure (Fig.1).

To avoid the need for a local power supply, the distribution point unit can be reverse powered from the customer premises over the existing copper wires. There is hence a power bottleneck at the DPU and consequently the need to minimize energy consumption. In general, to enable energy efficiency, G.fast standards define what is known as discontinuous operation (DO), which are specifically designed power saving modes that allow intermittent transmissions for different users.

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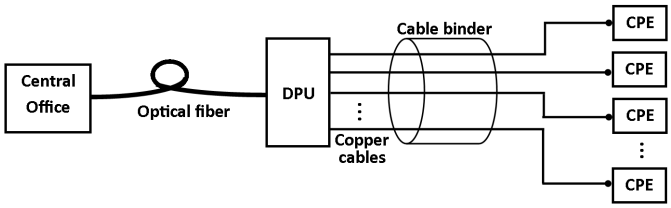

Fig. 1. G.fast FTTdp architecture.

Note that the G.fast transceivers use synchronized time division duplexing (TDD) and each TDD frame comprises of dedicated symbol positions (or time slots) for upstream and downstream data transmission. In the DO mode, front end of the transceivers can be disabled in the time slots they do not need to transmit data, thereby reducing the power consumption.

In the G.fast deployment model, copper wires connecting multiple users (lines) are grouped together in a binder. When users in the same binder transmit simultaneously, they interfere with each other causing crosstalk. Not canceling the crosstalk implies more slots are needed for transmission in order to satisfy data rate requirements for users, which results in higher power consumption. The approach used in traditional DSL systems for mitigation of crosstalk is to precode all the lines in a binder so that crosstalk can be canceled. However in G.fast, owing to discontinuous operation, all users need not be active in all time slots. Naively precoding all users when only a few are transmitting would lead to idle symbols being transmitted on the disabled lines, thereby sacrificing the power saving potential of DO. This context presents a new challenge in designing a precoder for crosstalk mitigation in G.fast.

A. Related Work

Maes and Nuzman in [1] were the first to discuss precoder design for DO in G.fast. They proposed a strategy called the *discontinuous vectoring (DV)* which exhibits reasonable power saving potential. It involves recomputing the input to the precoder at every time slot as the set of active users changes, and hence puts an extra computational burden on the system.

Motivated by the ideas around dynamic spectrum management (DSM) [2], another way to minimize the power consumption is to arrange users in DO slots such that they can avoid mutual crosstalk and/or minimize the usage of large sized precoders. Some of the earliest works in DO pattern design are by Zhang *et al.* [3] and Raj *et al.* [4]. In [3]

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authors group users into multiple non-overlapping vectoring sub-groups by proposing real-time algorithms to find an optimal grouping pattern to satisfy the given data demands. The biggest drawback of [3] is that it assumes that users are always precoded in the DO mode; it does not handle the unprecoded case at all. Even for the precoded case, it assumes that the rate of any user is independent of other users, which is a practically unjustifiable assumption. It will be useful to analyze whether precoding is necessary in the DO mode, which would make the aforesaid assumption invalid. Further, what is the trade-off between energy savings due to precoder and the complexity of estimation and application of precoders for all groups and time slots in DOI? Further, with or without the precoder, do overlapping groups in DO help in further energy savings? Finally, how does the general problem of user-slot allocation in the DO mode modify without the above assumptions? Our work attempts to answer the above questions.

B. Contributions

In this paper, we re-examine the energy efficiency problem for discontinuous operations in G.fast. Similar to [3], we treat this problem as an optimization problem wherein users are to be allocated time slots in a TDD frame. The objective is to minimize the overall power consumption while satisfying the constraints on the data rate of each user. In this paper, we tackle a general case wherein users may or may not be precoded in DO mode,[1] which implies that the data rate of a user in a time slot depends on other users in the same slot. This practical and general case opens up the combinatorial possibilities in the user-time slot allocation problem and makes the problem harder than the special case where user rates are independent of other users in the slot [3]. Moreover, our formulation works for any precoding strategy applied by the operator and also incorporates constraints on transmit opportunities and bit loadings of a user that are imposed by the G.fast standards [5] (more details in Sec II-B.1 and II-D).

The general user-time slot assignment problem is a combinatorial optimization problem and can be proved to be NP-hard. We propose an A^* based algorithm to obtain an optimal (or bounded sub-optimal) user-slot assignment in an efficient manner. The search efficiency of the A^* method crucially depends on the admissible heuristic, which we obtain in polynomial time by solving a relaxed version of the original problem using dynamic programming. For practical implementation and simulations we suggest weighted A^* [6] (using inflation factor ρ), which converts A^* to a (provably) bounded suboptimal algorithm [6] and takes significantly less computational time and memory with minimal compromise on the power efficiency. However, our simulations suggest that in most cases, the ρ value of one, implying optimality, will still result in reasonably fast computations.

The main insight that we derive is that the power budget when there is no precoder in the DOI may be the same as when precoder is applied in most scenarios. This is especially when overlapping groups (standard compliant) are allowed during DOI. This saves crucial computation needed to estimate precoders for every group and apply it during the DO mode.

In summary, our main contributions are:

1) We formulate the energy efficient discontinuous operations optimization problem with a comprehensive treatment of all possibilities.
2) Proved that the problem is NP hard in nature and provided an optimal algorithm that works quite fast in most practical scenarios.
3) Finally, provide a crucial insight that one does not even need to precode (which is an extremely complex operation) in most scenarios provided we perform a near optimal user-time slot allocation thus leading to practical utility of our work.

Notation: Bold upper- and lower-case letters denote matrices and column vectors. The $(i,j)^{th}$ element of matrix $\mathbf{A}$ is represented by $a_{i,j} = [\mathbf{A}]_{i,j}$. Superscript (k) associated with any quantity refers to tone index k; for example, $\mathbf{z}^{(k)}$ is the DMT signal for tone k.

II. System Model

A. G.Fast Architecture

Each G.fast subscriber is served via a dedicated twisted pair, and many such copper pairs are grouped together in a bundle. G.fast standards define two profiles with in-band spectral usage up to 106 MHz and 212 MHz respectively and uses DMT multi-carrier modulation. The 106 MHz profile consists of 2048 sub-carriers with a spacing of 51.75 kHz [5].

Unlike traditional DSL which uses Frequency Division Duplexing, G.fast uses Time Division Duplexing (TDD). A typical G.fast TDD frame consists of 36 DMT symbol positions, of which one is used as a guard interval, and the remaining are allocated to downstream and upstream data. Of these 35 symbols positions, those that may be used for data transmission form what is known as a Transmit Opportunity (TxOP). These include three types of symbols - *active* symbols during which actual data is transmitted, *idle* symbols during which no data is transmitted but the transceiver is still working, and *quiet* symbols during which the transceiver is turned off. A Dynamic Resource Allocation (DRA) engine controls the number of symbol positions allocated to each transmit direction - upstream and downstream [1].

B. Discontinuous Operation in G.Fast

To enhance energy efficiency, G.fast standards define Discontinuous operation [5]. Each logical frame (upstream or downstream) is divided into a normal operation interval (NOI) and a discontinuous operation interval (DOI). In the NOI, the transceivers of all lines in a binder are active at all times, as only active or idle symbols may be transmitted. The DOI, on the other hand, may consist of active, idle or quiet symbols. A DOI logical frame begins with zero or more quiet symbols, followed by a continuous stretch of data symbols depending upon the remaining rate requirement for the line, followed by quiet or idle symbols. Thus, at a given DOI symbol duration, some lines may be active while some may be discontinued (idle or quiet) (refer Fig.2).

[1]Even when users are precoded their rates may not be identical.

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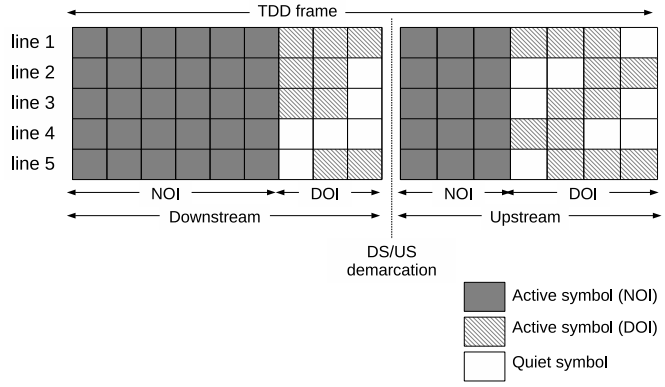

Fig. 2. A TDD frame comprises of downstream and upstream symbol positions. In each direction, a line remains active for some symbol positions and may go into discontinued mode when there is no data to transmit.

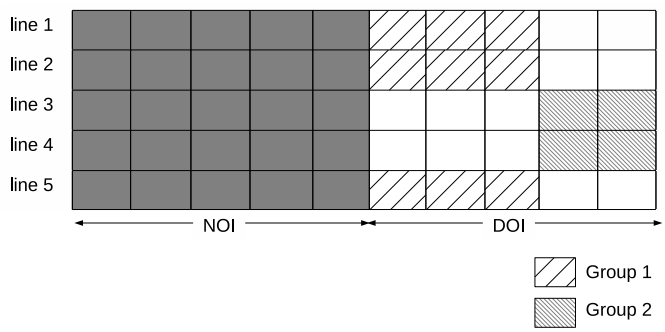

Fig. 3. Standard compliant grouped assignment: Each user in DOI only transmits with a fixed set of users.

1) DOI and G.Fast Standards: The G.fast standard [5] requires each user to transmit in contiguous time slots in the DO mode. In other words, during the DOI a user cannot transmit in intermittent time slots. Authors in [3] design a user-time slot assignment strategy such that the lines are subdivided into multiple small nonoverlapping subgroups during the DOI. While this is a valid configuration, the G.fast standards do not necessitate the formation of groups, and a DOI arrangement is valid as long as the active symbols on a given line are contiguous. In other words, the subgroups formed in the DOI may also overlap thereby giving us more degrees of freedom for the assignment problem. We denote the non-overlapping arrangement used in [3] as *grouped* and the overlapping case as *ungrouped*. Figures 3 and 4 illustrate the two cases of standard compliant assignment (grouped and ungrouped respectively).

C. Channel Model

Consider a G.fast binder comprising L lines, each serving one subscriber. Each line uses a set of subcarriers $k = 1, \ldots K$ for transmission over the crosstalk channel $\mathbf{H}^{(k)} \in \mathbb{C}_{L\times L}$. The system model for the kth tone is given by:

$$\mathbf{y}^{(k)} = \mathbf{H}^{(k)}\mathbf{z}^{(k)} + \mathbf{n}^{(k)} \tag{1}$$

where $\mathbf{z}^{(k)}$, $\mathbf{y}^{(k)}$ and $\mathbf{n}^{(k)}$ denote the users' DMT signal, received signal and additive noise at subcarrier k.

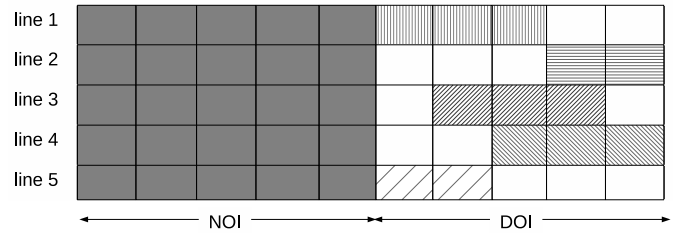

Fig. 4. Standard compliant ungrouped assignment: In the DOI, a user may transmit with different sets of users in different time slots.

D. Rate Calculations and Gain Scaling

Since users within a binder cause crosstalk to each other, vectoring for crosstalk cancellation in G.fast becomes essential. In this paper, we have used the diagonalizing precoder with column-norm gain scaling described in [1] to mitigate crosstalk while satisfying the PSD constraints. Note that our method is agnostic of the precoder and the corresponding gain scaling used. The system model with precoding, for the kth tone is given by (omitting superscript k for convenience):

$$\mathbf{y} = \mathbf{F}(\mathbf{H}\tilde{\mathbf{z}} + \mathbf{n}) = \mathbf{F}(\mathbf{H}\mathbf{Q}\mathbf{z} + \mathbf{n}) \tag{2}$$

where $\mathbf{z}$, $\tilde{\mathbf{z}}$, $\mathbf{y}$ and $\mathbf{n}$ denote the users' DMT signal, transmitted signal, received signal and additive noise respectively. $\mathbf{H}$ is the channel matrix, $\mathbf{Q}$ is the precoding matrix, and $\mathbf{F}$ is the diagonal equalizer at the receiver to compensate channel distortion. The precoder can be decomposed as $\mathbf{Q} = \mathbf{Q'S}$, where $\mathbf{Q'} = \mathbf{H^{-1}}$, and $\mathbf{S}$ is the scaling matrix that compensates for the power increase due to the precoder coefficients, computed as in [1].

The SNR for line l at subcarrier frequency k is calculated as [7]:

$$\gamma_l^{(k)} = \frac{p_l^{(k)} \left|h_{l,l}^{(k)}\right|^2}{\sum_{m\in\mathbb{M},m\neq l} p_m^{(k)} \left|h_{l,m}^{(k)}\right|^2 + N_l^{(k)}} \tag{3}$$

Here $p_l^{(k)}$ denotes the transmit power for line l at tone k, which is fixed according to the PSD mask for G.fast defined in the standard [5]. $N_l^{(k)}$ is the noise power for line l at tone k, $\mathbb{M}$ denotes the set of active users in a given time slot, and $h_{l,m}^{(k)}$ is the channel from line m to line l, i.e.

$$h_{l,m}^{(k)} = [\mathbf{H}^{(k)}]_{l,m} \quad \text{in the absence of precoder,} \tag{4a}$$

$$= [\mathbf{H^{(k)}} \times \mathbf{Q^{(k)}}]_{l,m} \quad \text{when precoder } \mathbf{Q} \text{ is used.} \tag{4b}$$

Given the SNR, the number of bits that can be reliably transmitted by line l at tone k is computed as follows [8]:

$$b_l^{(k)} \leq \min\left(\lfloor \log_2\left(1 + \frac{\gamma_l^{(k)}}{\Gamma}\right)\rfloor, b_{max}\right) \tag{5}$$

where Γ is the SNR gap computed as in [8], b_{max} is the maximum bit-loading per constellation and $\lfloor . \rfloor$ is the floor function.

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1) Remarks on Bit-Loadings and Gain Scaling: The G.fast standard requires a fixed set of bit-loadings for both the NOI and DOI for every user. During the NOI, SNR remains fixed for all symbol durations, so the bit-loading for each line can be assigned using (5) with equality. This however may not be true during the DOI. In our method, during the DOI, for every user, we consider the smallest value of the bit-loading for every tone and use the corresponding gain-scaling as well.

The total rate of user is simply given by:

$$R_l = \eta f_c \sum_{k=1}^{K} b_l^{(k)} \tag{6}$$

where η is the efficiency (includes transmission efficiency and coding overhead) and f_c is the sub-carrier spacing (51.75 kHz).

E. Energy Consumption Model

We use the energy consumption model described in [3]. Let T denote the total symbol positions in a data frame, and L the number of lines in a binder. Denote the length of NOI and DOI by T_{noi} and T_{doi} respectively. Let there be M unique active-user groups in the DOI. For group $m \in [1, ..., M]$, let u_m be the number of users in the group, and x_m the duration of the group.

There are two main sources of energy consumption in G.fast — data transmission and vectoring. During active and idle symbol periods, G.fast transceivers remais powered on, consuming energy. Energy due to data transmission thus scales linearly with the total number of active and idle symbols in a data frame, and is given by:

$$P_1 = \alpha_s \left(L T_{noi} + \sum_{m=1}^{M} u_m x_m \right) \tag{7}$$

where α_s represents the energy due to transmission of an active or idle symbol. In the quiet state, no power is consumed as transceivers are turned off.

The other major source of energy consumption is the precoder-data matrix multiplication during the vectoring step, which is given by:

$$P_2 = \alpha_c \left(4L^2 T_{noi} + \sum_{m=1}^{M} 4u_m^2 x_m \right) \tag{8}$$

where α_c is the multiplication energy consumption.

Then the total energy consumption is proportional to,

$$P = \left(L\, T_{noi} + \sum_{m=1}^{M} u_m x_m \right) + \beta \left(L^2\, T_{noi} + \sum_{m=1}^{M} u_m^2 x_m \right) \tag{9}$$

where $\beta = 4\alpha_c/\alpha_s$. The total power consumption P can also be written as summation of power in NOI and DOI. $P = P_{noi} + P_{doi}$.

$$P_{noi} = L\, T_{noi} + \beta L^2\, T_{noi} \tag{10}$$

$$P_{doi} = \sum_{m=1}^{M} u_m x_m + \beta \sum_{m=1}^{M} u_m^2 x_m \tag{11}$$

III. Proposed User-Slot Assignment Approach

Recall that the total energy defined in (9) is a function of number of active user symbols in both the normal mode (NOI) and the discontinuous mode (DOI). We frame an optimization problem to arrange the transmission of users (lines) across slots in both NOI and DOI such that total energy consumption P is minimized, subject to the constraint that each user meets its target data rate and the transit power constraints imposed by the G.fast standards. Additionally, the allocation of users to time slots in a frame should follow the G.fast standard specifications [5]. As discussed earlier, the standard requires each user to transmit in contiguous time slots in the DO mode. The other key compliance directive in G.fast is regarding the bit-loadings of users in the DOI. The G.fast standard mandates using a single bit-loading table for transmission in DO mode for every user. This implies that irrespective of SNR values of a user in different slots in the DO mode, the bit-loading of a user will only be derived by the minimum SNR that it achieves across all the DO time slots. Note that the SNR of users in the DO mode could vary in different time slots, unlike the NO mode where the SNR values do not change. Keeping the above constraints in mind, we now define the user-slot assignment optimization problem.

Note that the rate of user is function of the transit power in all tones as well as the transmit power of all the users in every tone. In other words, in an ideal scenario, the user-slot allocation problem is also a spectrum management problem. We begin by formulating the joint user-slot allocation as well as the power allocation problem. However, we notice that jointly solving user-slot allocation along with user spectrum management entails exorbitant complexity with limited rewards. We further notice that even if the transit power spectrum of each of the users is fixed, the remaining user-slot allocation problem is still NP complete, thereby proving that joint problem is NP hard as well.

A. Optimization Problem

Given T time slots and L users, we define a $L \times T$ association matrix $\mathbf{A}$ such that when user l transmits in slot t, $[\mathbf{A}]_{l,t} = 1$, otherwise 0 (refer Fig. 5). Also let $\mathbf{A}_t$ be the tth column of $\mathbf{A}$ and $R_l(\mathbf{A}_t)$ denote the rate of user l in time slot t. Clearly $R_l(\mathbf{A}_t) = 0$ if $[\mathbf{A}]_{l,t} = 0$. Let there be T_{noi} slots for the normal operations. Also, let d_l denote the slot index when the lth user begins transmission in the DO mode. Clearly,

$$T_{noi} + 1 \leq d_l \leq T \tag{12}$$

Let ζ_l denote the length of the discontinuous operations for user l. We collect the indices of transmission for user l in set

$$\chi_l = \{[1, \ldots, T_{noi}] \cup [d_l, \ldots, d_l + \zeta_l - 1]\} \tag{13}$$

Using the definition of matrix $\mathbf{A}$, we conclude that $[\mathbf{A}]_{l,t} = 1$ when $t \in \chi_l$. Note that one can generate all possible transmission strategies of all users by specifying T_{noi}, d_l and ζ_l and correspondingly fill the association matrix. Similarly,

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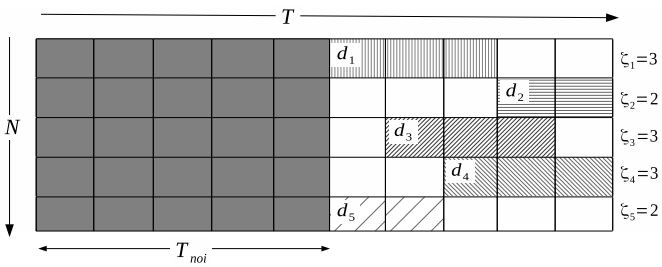

Fig. 5. $L \times T$ association matrix $\mathbf{A}$; blank cells represent a 0 (user not active).

let $\mathbf{P}$ be a $K \times L$ transit power matrix with $[\mathbf{P}]_{l,k} = p_l^{(k)}$. Thus, the user-slot assignment optimization problem is:

$$\begin{aligned} \textbf{OPT1:}\ & \underset{\mathbf{A},\mathbf{P}}{\text{minimize}}\, P \\ & \text{subject to } R_l^{total} \geq R_l^{target},\ l = 1, \ldots, L \\ & \quad (12), (13) \\ & \quad \sum_k p_l^{(k)} \leq p_l^{(total)}, \quad l = 1, \ldots, L \\ & \quad 0 \leq p_l^k \leq p_{mask}^k, \quad k = 1, \ldots, K \end{aligned} \tag{14}$$

where, R_l^{target} is the target rate and R_l^{total} is the total rate achieved by user l. Also, p_l^{total} is the transmit power budget and $p_{mask}^{(k)}$ is the mask. The last and the penultimate constraint define the spectrum constraints for the users. R_l^{total} is:

$$R_l^{total} = \sum_{t \in \chi_l} R_l(\mathbf{A}_t) = T_{noi}\, R_l^{noi} + \zeta_l\, R_l^{doi} \tag{15}$$

Notice that in (15) the total rate of the user is sum of its rate in NOI and DOI. The rate in NOI is straight forward and is given by (5) (with equality) and (6). The optimization variables of the above problem are the assignment matrix $\mathbf{A}$ and the transmit power matrix $\mathbf{P}$ as the total user rate is a function of both $\mathbf{A}$ and $\mathbf{P}$.

In the xDSL and G.fast literature the spectrum management problem is also known as Dynamic Spectrum Management (DSM) problem where the power spectral density of users are determined to minimize the interference between users (by optimizing a system wide utility function of user rates). Even for a single slot, that is when the index of users occupying that single slot are known, all the variants of DSM problems are NP-hard for more than two users and arbitrary number of tones as proved in the seminal work by Luo and Zhang [2].

We prove that given the PSD of users, the user-slot assignment problem is NP complete as well. Consequently, the joint user-slot allocation and the spectrum management problem (**OPT1**) is NP-hard. Since this work deals primarily with the user slot allocation problem, we simplify the problem by assuming that PSD of users are known apriori such that both the power budget as well as PSD mask constraints are satisfied. Consequently, the optimization problem **OPT1** assumes that the matrix $\mathbf{P}$ is known.

B. User Rate R_l^{doi} in DOI: Grouped and Ungrouped

1) Grouped: During the discontinuous operations slots, when the users are *grouped* into disjoint groups (Fig.3), the SNR of each user is identical across all active DOI slots. Since a user l belongs to a unique group, the SNR γ_l^k for l can be computed using the information of other users in the group (3), and the bit-loading b_l^k can be computed according to (5). Thus the rate for user l in DOI, R_l^{doi} in (15), can be calculated by:

$$R_l^{doi} = \eta f_c \sum_{k=1}^{K} b_l^{(k)},$$

Naturally, the SNR also depends on whether the group has been precoded or not.

2) Ungrouped: When users are not "grouped" during the DOI, SNR of a user may change due to the presence of different users in different slots. However, the standards do not allow the corresponding change in bit-loading. In this case, each user is assigned a DOI bit-loading corresponding to the worst-case SNR that is achieved in case any other user or set of users is active. The worst case SNR here corresponds to the minimum SNR over the current DOI allocation pattern, and not the minimum over all possible user combinations. Similarly, the worst case gain scaling is computed (corresponding to the minimum gain scaling across the time slots) to keep the allocation standard compliant. The simulation suggests that using the worst case gain scaling and separate gain scaling corresponding to each slot for each user has no impact in the final rate and power computations.

For precoded transmission, standardization only allows for a single gain scaling for DOI per user. In the grouped case, this is automatically enforced since each user belongs to a unique group, and thus uses only one precoder.

C. User-Slot Allocation and Role of Precoding

The assignment of users to time-slots is largely dependent on the rate at which a user transmits in a given time slot in the presence of other active users in that slot. Data rate is primarily a function of the SNR of the user, which depends on the precoder employed for crosstalk mitigation. This brings us to the question of whether a precoder should be used during the DO mode, and if yes, how should one precode? Is the "extra" cost of computing and applying the precoder worth the gain it provides in terms of data rate and energy consumption? An obvious approach is to precode all users in a time slot, but it leads to idle symbols being transmitted on the disabled lines, leading to additional energy consumption. On the other hand, one can precode only the set of active users in each time slot, a strategy used in [3]. This method, however, may not be implementation friendly since it calls for recomputing/applying the precoder during the DO slots. This will be specially burdensome when the user target rate changes after a few TDD frames. In this work, we work with the following precoding strategies in order to obtain data rate functions for the user-slot allocation problem.

- We test a simple precoding strategy, where we precode all users during the NO mode, and none in the DO mode. A major insight that we obtain is that with a good

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user-slot allocation, one can achieve the same energy consumption without even needing to precode in the DO mode.

- The other strategy is to precode all the users in the NOI, and only the active users in the DOI. This strategy is adopted by authors in [3].

It should also be noted that the proposed user-slot allocation approach is agnostic to the precoding scheme adopted.

The user slot assignment problem $OPT1$ in (14) is NP hard. Note that it is a G.fast compliant formulation where the users are only allowed to transmit in contiguous time slots. To prove the NP hardness of $OPT1$, we construct a corresponding feasibility problem and show that even obtaining feasible user slot assignment is NP hard.

Theorem 1: For a given value of T_{noi}, the user-slot assignment problem $OPT1$ (14) is NP hard.

Proof: The proof is carried out in the appendix. ■

IV. Algorithm

Recall that the power minimization problem for G.fast (14) is essentially a combinatorial optimization problem of finding an optimal user slot assignment matrix $\mathbf{A}$ and is NP hard in general. We now describe a weighted A^* based algorithm to solve the user-slot assignment problem.

1) Motivation: The A^* algorithm [9] explores the search space in a systematic best first manner, using a heuristic to guide the search. A systematic search approach of A^* provides some advantages over other search methods. First, it can guarantee an optimal solution which is generally not possible by non-systematic algorithms like genetic algorithm or tabu search. Second, it also guarantees that no other optimal algorithm can improve upon its performance (in terms of nodes expanded). Also, A^* can be easily modified to an approximate algorithm (weighted A^*) using a weighted heuristic; weighted A^* guarantees bounded sub optimal termination and is generally orders of magnitude faster than A^*, which makes it very useful for solving large scale minimization problems within reasonable resource constraints. Finally, the computational efficiency of the A^* algorithm depends on the heuristic employed, which as we shall see later is chosen in a manner that makes our overall search efficient.

The search space for the optimization problem (14) is the set of all feasible ways of filling the $L \times T$ association matrix $\mathbf{A}$ with 1's and 0's subject to the grouping constraints in DOI. The structure of $\mathbf{A}$ provides a crucial insight for reducing the search space for the A^* algorithm. Recall that for any valid allocation, the assignment matrix $\mathbf{A}$ can be completely specified by T_{noi}, d_l and ζ_l for all users $l \in [1, L]$. If the NOI duration is fixed, the first T_{noi} columns of $\mathbf{A}$ are already assigned a vector of all ones, and the problem reduces to filling the remaining columns of the matrix ($T_{noi} + 1$ to T), which can be uniquely specified by d_l and ζ_l, $\forall l \in [1, L]$; see (12) and (13). We thus take a two-step approach to solve (14). We first fix the NOI duration in allocation matrix $\mathbf{A}$, and use the A^* algorithm to find allocation for the DOI time slots. We perform DOI allocation for each permissible value of T_{noi} ($3 \leq T_{noi} \leq T$) and choose the allocation matrix corresponding to that value of T_{noi} which minimizes energy P. Suppose the rate obtained by user l in one NO slot is R_l^{noi}, then the remaining rate requirement that needs to be satisfied for user l using the DOI is $R_l^{target} - T_{noi}\, R_l^{noi}$. The allocation matrix $\mathbf{A}$ can be written as $[\mathbf{A}_{noi} \mid \mathbf{A}_{doi}]$, where, $\mathbf{A}_{noi}$ denotes the first T_{noi} columns of $\mathbf{A}$ that are filled with ones. The modified DOI allocation optimization problem that minimizes the energy consumption P_{doi} given T_{noi} is

***Problem 1*:**

$$\begin{aligned} &\underset{\mathbf{A}_{doi}}{\text{minimize}} \;\; P_{doi} \\ &\text{subject to } \zeta_l\, R_l^{doi} \geq R_l^{target} - T_{noi}\, R_l^{noi}, \quad l = 1, \ldots, L \\ &\qquad (12), (13), \end{aligned} \tag{16}$$

$$P_{doi} = \left(\sum_{m=1}^{M} u_m x_m \right) + \beta \left(\sum_{m=1}^{M} u_m^2 x_m \right) \tag{17}$$

2) Remarks:

- The rate obtained by a user in one DO slot R_l^{doi} depends on the other users in the same slot which in turn depends on the precoding strategy that is employed. Our formulation is agnostic of above settings and works for all possible scenarios. For instance, when a precoder is not employed in DOI ($\beta = 0$ in the calculation of P_{doi}), users are arranged such that crosstalk is avoided.
- Even though it is suggested to run the DOI allocation algorithm for each permissible value of T_{noi}, in practice our algorithm only runs for a few values. This is because for higher values of T_{noi}, the rate constraints may already be satisfied using NOI and hence there is no need to perform DOI allocation. Similarly, for lower of values of T_{noi}, the DOI allocation problem may become infeasible as the remaining DO slots may not be able to satisfy the rate constraints. The feasibility check can be performed using a lower bound on the DOI allocation cost, as described in Sec IV-B.

The complete approach is summarized in Algorithm 1.

A. Weighted A^* Based Search for DOI Allocation

We map Problem 1 to a graph search problem [9], where a *node* denotes a partially or completely filled association matrix $\mathbf{A}_{doi}$. In particular, a node q with τ filled columns represents a partial user-slot assignment in which τ DOI time slots have been allocated so far (note that there are $2^{(L-1)\tau}$ ways of doing so; one node represents only one such way), while the remaining are yet to be assigned. A *successor* q_s of node q is any node obtained by assigning users to the first unfilled column of q. The cost associated with the path from q to q_s is the energy consumption for the newly filled column according to (17). We define the *start node* q_0 as an empty allocation matrix, and a *goal node* as a matrix that satisfies the target rates of all users. The objective is to find the minimum cost path from the start node to a goal node.

The A^* algorithm defines an evaluation function for each node q. Let $g(q)$ denote the cost of getting to q from the start node q_0, which in our case is the energy consumption for the τ filled columns in DOI. Corresponding to the assignments in the first τ columns, each user achieves certain data rate. If the total

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Algorithm 1 Framework for Solving (14)

Input: L users, T time slots, $R_l^{target}, b_l^{(k)} \quad \forall l \in [1, L], k \in [1, K]$
Output: association matrix $\mathbf{A}^*$
for $T_{noi} = T$ to 0 **do**
  Initialize $\mathbf{A}(T_{noi}) = [\mathbf{A}_{noi} \,|\, \mathbf{A}_{doi}]$
  where $\mathbf{A}_{noi}$ denotes the first T_{noi} filled columns of $\mathbf{A}$, and $\mathbf{A}_{doi}$ are the remaining unfilled columns
  Calculate the rate obtained in NOI in one slot, $R_l^{noi} \quad \forall l$
  DOI rate required $\leftarrow R_l^{target} - T_{noi} \times R_l^{noi}$
  if DOI rate required is greater than 0 for any user i **then**
    perform DOI allocation using weighted-A^*
    if solution is obtained from weighted-A^* **then**
      $\mathbf{A}_{doi} \leftarrow$ solution obtained from weighted-A^*
    end if
  end if
  Calculate energy consumption $P(T_{noi})$ from $\mathbf{A}(T_{noi})$ using (9)
end for
$T_{noi}^* = \arg\min P(T_{noi})$
$\mathbf{A}^* = \mathbf{A}(T_{noi}^*)$

target rates of the users are not achieved, further assignments need to be made to the remaining time slots. We define $h(q)$, also called a heuristic, as a pointer to the cost to the goal node from q, *i.e.* an estimate of the minimum energy consumption $h^*(q)$ of the remaining assignments at node q. A heuristic is said to be *admissible* if $h(q) \leq h^*(q)$ or in other words $h(q)$ is a lower bound on the optimal energy consumption on the remaining assignments. A method for obtaining an admissible $h(q)$ is discussed later in Sec IV-B. Finally, the evaluation function $f(q)$ for node q is defined as follows

$$f(q) = g(q) + \rho\, h(q) \tag{18}$$

where ρ is an inflation factor ($\rho \geq 1$) that bounds the sub-optimality of the solution.

The search maintains a sorted *OPEN* list which is ordered by the evaluation function in (18). *OPEN* initially consists only of the start node q_0. The search begins with the start node, expanding nodes as it progresses. In every *expansion*, the node with the lowest f-value is removed from *OPEN*, all possible successors of the node are generated and inserted in *OPEN*. When the node with the lowest f-value in *OPEN* is a goal node, the search halts and the incumbent node is returned as the solution. [4] provides a detailed illustration of the method.

1) Remark on Feasibility: As discussed earlier, the G.fast standards compliant allocation matrix must adhere to certain grouping constraints in the DOI. To ensure that the weighted-A^* algorithm only searches through nodes that are permissible by standards, we must check for the feasibility of the allocation pattern every time a new node is inserted in the *OPEN* list. For instance, once a user has stopped transmission in the DOI, it is not allowed to be further present in future node expansions.

Further, recall that users can be arranged in grouped and ungrouped manner. Depending on the type of arrangement scheme adopted, we check whether a new node confirms to the respective arrangement criteria or not. In the ungrouped arrangement, because of the constraints mentioned in Sec III-B.2, at every possible allocation, we change the rates that have been obtained previously based on the subsequent choices in the later slots while honoring the constraints. And in this process, since we search all possible combinations and re-compute the previous rates as well, we do not encounter any suboptimality.

B. Admissible Heuristic $h(q)$

An admissible heuristic in the A^* algorithm is a lower bound on the cost of optimal assignment for the remaining and unfilled DO slots. A heuristic which achieves a tight lower bound leads to a more efficient search. To achieve a lower bound on energy (for remaining user rate requirements), we relax the optimization problem (16) in the following ways.

- We replace individual user rate constraints by one sum rate constraint.
- We redefine the sum rate of active users in a DO time slot by replacing it with an upper bound, thereby potentially needing fewer slots to achieve the target sum rate.
- Finally, we also replace the objective function in (16) with its lower bound.

With these relaxations, the resulting optimization problem can now be solved in polynomial time via a dynamic programming technique to obtain a heuristic.

Consider node q with τ filled DOI columns. Let $\mathbf{A}_{doi}(q)$ represent the partially filled DOI allocation matrix corresponding to node q (for time slots $T_{noi} + 1 \leq t \leq T_{noi} + \tau$), and $\tilde{\mathbf{A}}_{doi}(q)$ denote the remaining unfilled columns of $\mathbf{A}$ ($T_{noi} + \tau + 1 \leq t \leq T$). This means

$$\mathbf{A}(q) = \left[\mathbf{A}_{noi} \,|\mathbf{A}_{doi}(q) \,|\, \tilde{\mathbf{A}}_{doi}(q)\right] \tag{19}$$

Each user achieves certain data rate in the initial $T_{noi}+\tau$ filled columns. Let $\tilde{R}_l^{target}(q)$ denote the remaining rate requirement of user l. Naturally, $0 \leq \tilde{R}_l^{target}(q) \leq R_l^{target}$.

We define a quantity $R_{sum}(n)$, which denotes the maximum sum-rate that can be obtained by any combination of n active users in a time slot.

$$R_{sum}(n) := \max_{j:|\mathbf{A}_j|=n} \sum_{l=1}^{L} (R_l(\mathbf{A}_j)) \tag{20}$$

To avoid looping through all n-tuple user combinations, approximations to $R_{sum}(n)$ can be used. Details are in Sec IV-D.

We solve the following problem to obtain heuristic $h(q)$:

***Problem 2*:**

$$\begin{aligned} \underset{\tilde{\mathbf{A}}_{doi}(q)}{\text{minimize}} \;\; & \tilde{\tilde{P}}_{doi}(q) = \sum_{t=T_{noi}+\tau+1}^{T} |\mathbf{A}_t| \\ \text{subject to} \;\; & \sum_{t=T_{noi}+\tau+1}^{T} R_{sum}(|\mathbf{A}_t|) \geq \sum_{l=1}^{L} \tilde{R}_l^{target}(q) \end{aligned} \tag{21}$$

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The objective function here is same as the sum of entries of $\tilde{\mathbf{A}}_{doi}(q)$. Problem-2 essentially means that the sum-rate in a time slot depends on the number of users active in the slot; and the objective is to decide the number of active users in each time slot of $\tilde{\mathbf{A}}_{doi}(q)$ such that the total rate achieved across all users is no less than the sum of target rates of all users. It is easy to see that if $\tilde{P}_{doi}(q)$ represents the energy consumption in $\tilde{\mathbf{A}}_{doi}(q)$, as given in (11), then $\tilde{\tilde{P}}_{doi}(q) \leq \tilde{P}_{doi}(q)$, and therefore this would lead to the required lower bound.

Lemma 1: If an admissible heuristic is used, the solution returned by weighted A^* is within a factor of ρ of optimal.

Proof: Please refer to [6] for proof. ■

Theorem 2: $h(q)$ obtained from solving problem 2 is an admissible heuristic.

Proof: We now show that the solution to optimization problem (21) *provides a lower bound to the solution of the original user to slots problem* (16) *evaluated at node q. Let Q denote the constraint set in* (16) *at node q. And let $Q^{'}$ denote the constraint set in* (21)*. We rewrite the optimization problem* (16) *at node q in the following manner:*

$$\begin{aligned} &\underset{x}{\text{minimize}} \;\; f(x) \\ &\text{subject to } x \in Q \end{aligned} \tag{22}$$

Similarly, we rewrite (21) *as*

$$\begin{aligned} &\underset{x}{\text{minimize}} \;\; g(x) \\ &\text{subject to } x \in Q' \end{aligned} \tag{23}$$

Since we have relaxed the individual rate constraints to sum rate constraints, $Q \subseteq Q^{'}$. Note that $f(x)$ is the total power dissipation, while $g(x)$ is the power dissipation when $\beta = 0$ (17)*. Therefore, $g(x)$ lower bounds $f(x)$. Let the optimal solution to* (22) *and* (23) *be x^* and $x^{'*}$ respectively. The above two statement implies $g(x^*) \leq f(x^*)$. But since $x^{'*}$ optimizes* (23)*, we have $g(x^{'*}) \leq g(x^*)$. Hence $g(x^{'*}) \leq f(x^*)$. In words, $h(q) = g(x^{'*})$ acts as an admissible heuristic.* ■

C. Dynamic Programming Method to Obtain $h(q)$

We now look for an efficient method to solve (21). For a node q with τ filled DOI slots, $\tilde{R}_l^{target}(q)$ denotes the minimum sum-rate that must be achieved in the remaining $(T - T_{noi} - \tau)$ time slots by user l. The sum rate (or gain) achieved with n active users in one slot is given by $R_{sum}(n)$. As noted earlier, the objective function in (21) is to determine the binary allocation matrix $\tilde{\mathbf{A}}_{doi}(q)$ with total number of ones (cost of allocation) minimized while satisfying a minimum target rate.

To find the minimum cost, we first solve a related problem of finding maximum sum-rate that can be achieved given a fixed cost c, $\forall c \in [1, (L-1) \times (T - T_{noi} - \tau)]$. The minimum value of c at which sum-rate $\sum_l \tilde{R}_l^{target}(q)$ is obtained in $(T - T_{noi} - \tau)$ slots is the solution cost, and the corresponding $\tilde{\tilde{P}}_{doi}(q)$ gives the desired heuristic $h(q)$. The problem of determining the maximum achievable sum-rate in $(T - T_{noi} - \tau)$ time slots given a cost c exhibits optimal substructure - one can build an optimal solution to the original problem from the optimal solution of sub-problems. In other words, one can apply Dynamic programming to solve Problem 2.

Lemma 2: Suppose we know that optimal solution of c-cost problem has n active users in the $(T_{noi} + \tau + 1)^{th}$ slot, then the optimal solution consists of an optimal solution to the remaining $(c - n)$ cost problem in $(T - T_{noi} - \tau - 1)$ slots, plus n active users in the $(T_{noi} + \tau + 1)^{th}$ slot [10].

Let $\Delta R(T - T_{noi} - \tau, c)$ denote the maximum sum-rate that can be achieved in $T - T_{noi} - \tau$ slots, given cost c. Using Lemma 2, it can be obtained with the help of the following recursion

$$\Delta R(T - T_{noi} - \tau, c) = \max_{n \in [0, L-1]} \{\Delta R(T - T_{noi} - \tau - 1, c - n) + R_{sum}(n)\} \tag{24}$$

Through the recursion (24), one can reach the optimal $\Delta R(T - T_{noi} - \tau, c)$ using optimal $\Delta R(T - T_{noi} - \tau - 1, c - n)$ by searching over all possible number of active users ($n \in 0, \ldots, L-1$) in the $(T_{noi} + \tau + 1)^{th}$ slot. The runtime of the dynamic algorithm is $O(LT^2)$ and is run just once to determine the maximum sum rate (gain) that can be achieved for each of the values of cost c ranging from 1 to $(L-1)T$.

D. Note on Computational Complexity

The complexity and speed of the weighted A^*-based search is dependent on the inflation factor ρ. The search is optimal for $\rho = 1$. Higher values of ρ lead to more suboptimality (bounded) but would be computationally faster [6]. In our experiments, majority of the reported results have been obtained using $\rho = 1$. However, in case a solution is not obtained within a set limit on the number of node expansions, we halt the current search and re-do it using progressively higher values of ρ. For higher values of L, where each node expansion involves more operations, the limit on node expansions can be lowered. The search can thus be easily customized based on the computational constraints and optimality requirements.

The computation on $R_{sum}(n)$ in (20) requires evaluating 2^L user-combinations for L users. This is a one-time computation for a binder and can be easily computed for $L = 10$ users. For higher values of L, computation of all 2^L rate-combinations can be avoided by using an upper bound on $R_{sum}(n)$ without affecting the admissibility of the resulting heuristic.

To obtain an upper bound on $R_{sum}(n)$, we start with an upper bound on the capacity of communication to the lth CPE (obtained when transmissions from each of the L transmitters are phase-aligned and add coherently at the CPE) on tone k, which is given as:

$$c_l^{(k)} = \min\left(\left\lfloor \log_2\left(1 + \frac{\sum_{j=1}^{L}\left(|h_{l,j}^{(k)}|\sqrt{p_j^{(k)}}\right)^2}{\Gamma N_l^{(k)}}\right)\right\rfloor, b_{max}\right) \tag{25}$$

The maximum achievable rate for user l across all tones, C_l, is calculated similarly to (6). The upper bound on the sum-rate

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of n active users, $R_{sum}(n)$, is then computed as the sum of the highest n channel capacities among all users.

$$R_{sum}(n) = \max_{S:|S|=n} \sum_{l \in S} C_l \tag{26}$$

where S is a subset of the L users in the binder, with any n users active. Since this gives a loose upper bound, we calculate the exact $R_{sum}(n)$ for a few values of n, say $n \in \{1, 2, 3\} \cup \{L-2, L-1, L\}$. This brings down the complexity of (20) from $O(2^L)$ to $O(L^2)$. Our experiments reveal that even with this upper bound, good heuristics are obtained and the run-time of weighted-A^* search is not hampered (refer Sec V-C).

V. Simulation Results

We now evaluate the performance of the proposed user-slot assignment based power minimization method and compare with other existing power minimization algorithms for G.fast. The simulation parameters are chosen to reflect realistic traffic conditions. We have considered: efficiency $\eta = 0.85$, SNR gap $\Gamma = 10.75$ dB, and maximum bits per bin $b_{max} = 14$.

A. Setup

For the simulations, we consider three G.fast binder topologies: (i) **Binder-1** comprising 10 pairs of equal loop lengths (100m each), (ii) **Binder-2** comprising 10 pairs with mixed loop lengths (varying from 100m to 300m) and (iii) **Binder-3** comprising 20 pairs of equal loop lengths (100m each). Crosstalk data for all binders has been obtained from channel measurements. Data for Binders 1 and 3 has been provided by British Telecom, while data for Binder-2 has been provided by Swisscom. All sets of measurements are for the 106 MHz profile. The transmit PSD from the G.fast standard for the 106 MHz profile is used. Background noise is AWGN with a PSD of -140 dBm/Hz.

To simulate the demands for each frame, we consider two Traffic Patterns following [1]. **TP1** is a heavy usage traffic pattern where all users of a binder have identical rate requirements, ranging from 200 Mb/s to the highest rate achievable with vectoring. **TP2** on the other hand, is a light usage pattern. Only two users in a binder (the ones with the highest and lowest aggregate channel capacity) have varying target rates, whereas all other users have a fixed target rate of 200 Mb/s. Note that 200 Mb/s is the minimum data rate that G.fast systems are expected to sustain [11].

B. Comparison of Energy Efficiency

Given a set of target rates to be met in a data frame, we allocate users to time slots using the algorithm proposed in Sec IV, and report the energy consumption for the frame (calculated using (9)). In Sec III, we have discussed two ways of grouping users across a TDD frame, and two precoding schemes, which lead to the following four arrangements:

(a) *Grouped-Precoded*: This is the setting for the algorithm proposed by Zhang *et al.* [3], which we replicate using our weighted-A^* based method. Here, the DOI comprises of non-overlapping user groups, and each group is precoded.

(b) *Grouped-Not-Precoded*: This scheme has similar arrangement of users in the DOI as the previous scheme, but we propose not to precode during the DOI.

(c) *Ungrouped-Precoded*: Here we relax the constraint of non-overlapping groups in the DOI, and allow a user to transmit with different sets of users in different time slots. All resulting groups are precoded.

(d) *Ungrouped-Not-Precoded*: This is similar to the previous scheme, but does not employ a precoder in DOI.

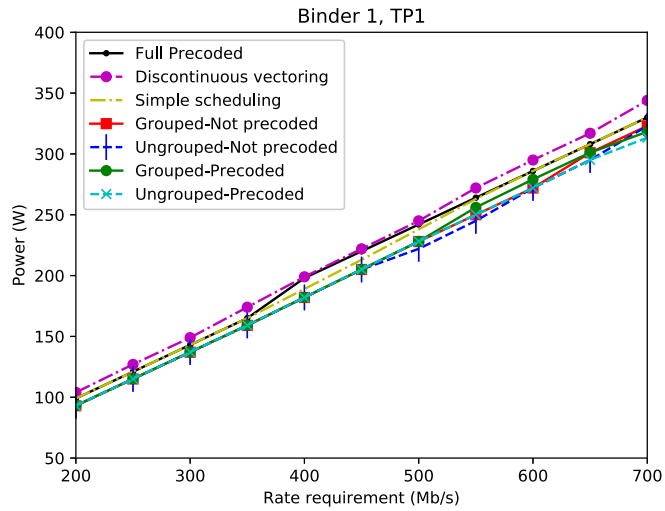

Fig. 6. Energy consumption for Binder 1 subject to Traffic Pattern 1.

We compare the energy efficiency of the proposed methods – (b), (c) and (d), with Zhang *et al.*'s (a) and with three other algorithms, *viz.*:

(e) *Full Vectoring*: All users are in a common vectoring group resulting in transmission only in NOI.

(f) *Discontinuous Vectoring*: This is Maes et al.'s method of using virtual precoder input to minimize idle symbols [1].

(g) *Simple Scheduling*: This is an easily computable scheduling approach where users are not precoded in the DOI. For a given NOI duration, the length of DOI is computed independently for each user based on the target rates. The duration of NOI is chosen by looping over permissible values, so as to minimize the total energy consumption.

Figures 6-9 show the energy consumption of a binder (for all users over a TDD frame) as a function of the target data rate, for binders 1 & 2 under different traffic patterns. Energy consumption is calculated using (9), with $\beta = 0.01$. Fig 6 presents a very simple scenario where all users have equal target rates, and nearly equal channel capacities. All methods perform roughly similar in this case. For the other figures, it can be seen that energy consumption for Full Vectoring is the highest because precoding all users leads to idle symbols being transmitted on the discontinued lines. Discontinuous Vectoring avoids transmitting idle symbols and hence the resulting energy consumption is much lower than that of Full Vectoring (in Fig 6 residual noise due to Discontinuous Vectoring leads to lower rates and hence higher energy consumption compared

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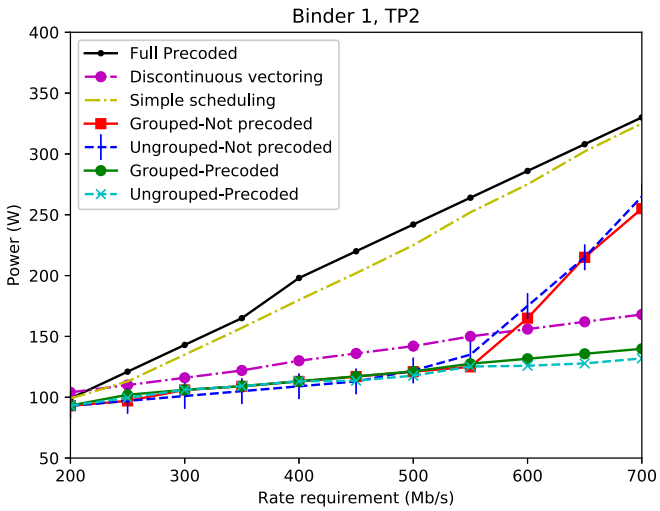

Fig. 7. Energy consumption for Binder 1 subject to Traffic Pattern 2.

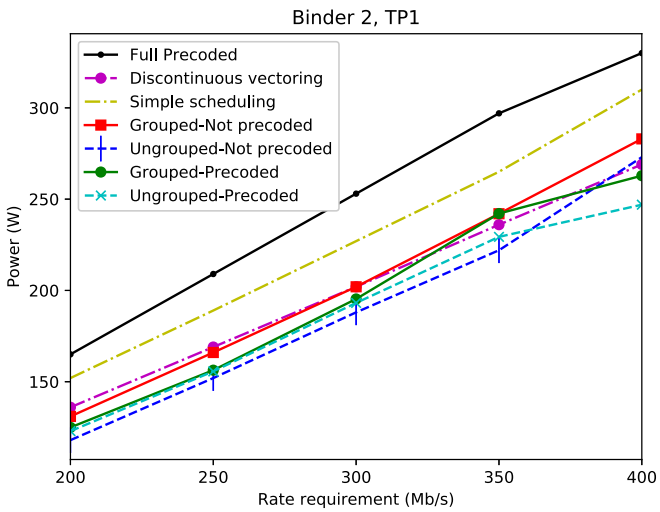

Fig. 8. Energy consumption for Binder 2 subject to Traffic Pattern 1.

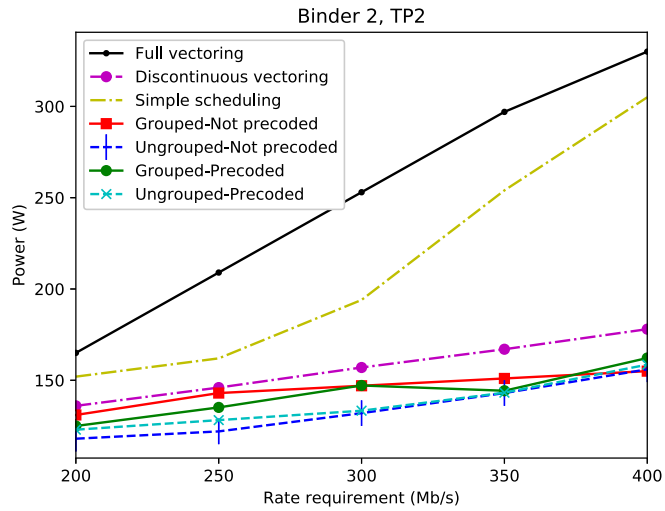

Fig. 9. Energy consumption for Binder 2 subject to Traffic Pattern 2.

TABLE I
COMPLEXITY: NO. OF STATE EXPANSIONS

Target rate (Mbps)	10 users		20 users	
	Exact sum-rate	Upper bound	Exact sum-rate	Upper bound
200	130	34	2483	2483
300	40	8	15575	15573
400	181	245	13536	13489
500	24006	24002	13354	13376
600	24005	24015	17504	17507
700	23792	24947	23428	26735

to Full Vectoring). The Simple Scheduling approach has energy consumption lower that Full Vectoring but much higher than the other proposed schemes or Discontinuous Vectoring. Energy consumption for the Grouped-Precoded strategy is much lower than Full Vectoring or Simple Scheduling.

Energy consumption for the proposed methods ((b), (c), (d)) is at many points lower than Zhang et al.'s (a). In particular, we can infer the following from the plots:

(i) Allowing transmission in overlapping groups in DOI leads to lower energy consumption compared to non-overlapping groups. In all the plots, energy consumption of Ungrouped-Precoded is lower than that of Grouped-Precoded; similarly, energy consumption of Ungrouped-Not-Precoded is lower than Grouped-Not-Precoded.

(ii) For low target rates, it may be advantageous to not precode at all during the DOI. In Figures 6 and 9, the (Un)Grouped-Not-Precoded scheme has equal or even lower energy consumption than (Un)Grouped-Precoded. However, for higher target rates in Figures 7 and 8, which is a case of large disparity in the users' demands, the absence of a precoder does lead to a predictable increase in energy consumption.

C. Complexity Analysis

To provide an idea of the practical complexity of the proposed approach, Table I presents the total number of state searches performed by Algorithm 1 for the Grouped-Precoded method, under different target rates (TP1) for Binders 1 and 3. As can be noticed that for both 10 and 20 users case, the average total number of state expansions are not high and can be easily computed in real-time in a standard processor. Table I also demonstrates the impact of using an upper bound to $R_{sum}(n)$ on the number of state expansions, as proposed in Sec IV-D. It can be seen that the upper bound causes no noticeable change in the complexity of search.

VI. CONCLUSION

We investigate the energy efficient algorithms for DO in G.fast. The user to slot assignment is a crucial element in the process for carbon footprint reduction of the G.fast devices. We discuss a discrete optimization based approach to assigning users to slots so as to minimize the energy consumption.

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We illustrate that in most cases, when near optimal user-time slot allocation is performed, precoding only in the normal operation interval is practically required and precoding in the discontinuous operation interval does not yield significant energy savings.

APPENDIX

A. Proof of Theorem 1

Problem K_COLOR: Given a graph $G = (V, E)$, the k-coloring problem requires the determination of whether or not there is a partition of the vertex set V or $V = V_1 \cup V_2 \cdots \cup V_k$ such that $V_1, V_2 \ldots V_k$ are all independent sets.

Problem DEC_OPT1: Let $S \subset [1, L]$ denotes the set of users transmitting in a time slot. Let the rate gain function in a time slot for the ith user be $R_i(S) : 2^L \rightarrow \mathbb{R}^+$. Also let the rate threshold for the ith user be R_i^{th}. Problem DEC_OPT1 is following. Given the rate gain function, rate threshold R_i^{th}, $\forall i \in [1, L]$ and the total number of available slots T, does there exist a feasible assignment of users to time slots such that the individual rate constraint of each user is satisfied.

Lemma 3: DEC_OPT1 is NP hard.

Proof: To prove NP-hardness of DEC_OPT1 we construct a polynomial time reduction from K_COLOR to DEC_OPT1. For an arbitrary graph $G = (V, E)$ we define an instance of DEC_OPT1 named $\mathbf{I}$ as follows. For every vertex $v \in V$, we construct a user v and its rate function as follows

$$R_v(S) = \begin{cases} 1, & \text{if } v \in S \text{ and } (u, v) \notin E, \ \forall u \in S \setminus \{v\} \\ 0, & \text{otherwise} \end{cases} \tag{27}$$

The rate threshold $R_v^{th} = 1$. We show that the graph G is k-colorable iff $\mathbf{I}$ is an Yes instance with parameter $T = k$.

Let vertex sets $V_1, V_2, \ldots V_k$ be independent sets constituting the k-coloring of G. We assign time slots to users in the following manner. For all vertices $v_j \in V_i$, we assign user index j the ith time slot. This assignment operation is performed for all sets V_i, $1 \leq i \leq k$. This is a feasible assignment as all users in a particular time slot achieve a gain of one since vertex sets V_i are independent sets. Clearly $T = k$, hence $\mathbf{I}$ is an Yes instance.

Suppose we have the solution for $\mathbf{I}$. Let $\mathbf{M}$ denote the corresponding user-time slot assignment matrix of dimension $|V| \times T$. We have the following assignment.

$$V_i = \{v_j \mid R_j(\mathbf{M}_i) = 1, 1 \leq j \leq |V|\}, \quad 1 \leq i \leq T \tag{28}$$

where, $\mathbf{M}_i$ denotes the ith column of matrix $\mathbf{M}$. Since set V_i contains only users with gain one, the set of users V_i form an independent set. Further since every user satisfies the requirement of gain one, vertex corresponding to every user is part of exactly one independent set. This proves that the problem DEC_OPT1 is NP hard. ■

Since DEC_OPT1 is a decision version of $OPT1$, therefore $OPT1$ is also NP hard.

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